

Customer Segmentation & Churn Pattern Analytics in European Banking

Prepared for: The European Central Bank (via Unified Mentor) **Analysis type:** Exploratory Data Analysis, Segmentation Design, Churn Pattern Analytics **Dataset:** 10,000 retail banking customers across France, Spain, and Germany

Abstract

Customer churn is one of the largest hidden costs in retail banking, driving down lifetime value, raising acquisition costs, and destabilizing revenue. This paper presents a segmentation-driven analysis of a 10,000-customer dataset from a European retail bank operating in France, Spain, and Germany. We validate and clean the data, construct five segmentation dimensions (geography, age, credit score, tenure, and balance), and quantify churn behavior across each. We find an overall churn rate of **20.37%**, with pronounced concentration among customers aged 46–60, customers holding three or more products, inactive members, and — critically — high-value customers, who churn at a higher rate (25.1%) than the customer base as a whole. We close with segment-specific retention recommendations.

1. Background and Context

Banks routinely track an aggregate churn rate, but an aggregate number cannot answer the questions that matter operationally: *which* customers are leaving, *where*, and *why*. Without granular segmentation, retention strategies default to blanket, reactive campaigns that spend budget on customers who were never going to leave while missing the segments driving most of the loss. This project addresses that gap by building a structured, segmentation-driven analytics layer on top of customer-level banking data.

2. Problem Statement

Despite having rich customer-level records, the bank faces three concrete challenges:

1. Identifying which customer segments carry disproportionate churn risk.
2. Understanding how churn differs across the three countries and across demographic groups within them.
3. Quantifying the financial exposure — deposit balances and income — tied to the customers who actually leave.

3. Dataset Description

The dataset contains 10,000 customer records with the following fields: `CustomerId`, `Surname` (dropped — not analytical), `CreditScore`, `Geography` (France, Spain, Germany), `Gender`, `Age`, `Tenure` (years with the bank), `Balance`, `NumOfProducts`, `HasCrCard`,

IsActiveMember, EstimatedSalary, and the binary target Exited. The data contained no missing values, and all binary fields (HasCrCard, IsActiveMember, Exited) were confirmed to contain only 0/1 values during validation.

Class balance: 79.63% retained vs. 20.37% churned — a moderate imbalance that is accounted for by reporting rates and shares rather than raw counts throughout this analysis.

4. Methodology

4.1 Data Ingestion & Validation

The raw CSV was loaded and checked for null values, valid category labels (Geography, Gender), and binary-field consistency before any analysis began.

4.2 Data Cleaning & Preparation

The non-analytical Surname field was removed, and the data was checked for duplicate CustomerId records (none were found).

4.3 Segmentation Design

Five segmentation dimensions were engineered from the raw fields:

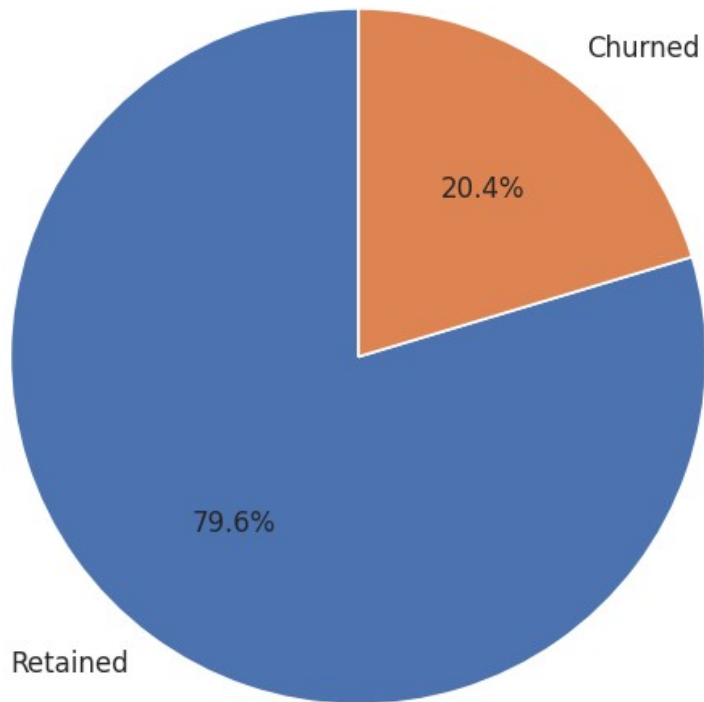
Dimension	Bands
Geography	France, Spain, Germany
Age	<30, 30–45, 46–60, 60+
Credit Score	Low (≤ 580), Medium (581–700), High (701+)
Tenure	New (0–2 yrs), Mid-term (3–6 yrs), Long-term (7+ yrs)
Balance	Zero-balance, Low-balance ($< \text{€}100\text{k}$), High-balance ($\geq \text{€}100\text{k}$)

A **High-Value** flag was additionally derived for customers whose balance *and* estimated salary are both at or above the dataset median, and an **Engagement Level** field combining activity status and product count.

5. Churn Distribution Analysis

5.1 Overall Churn Rate

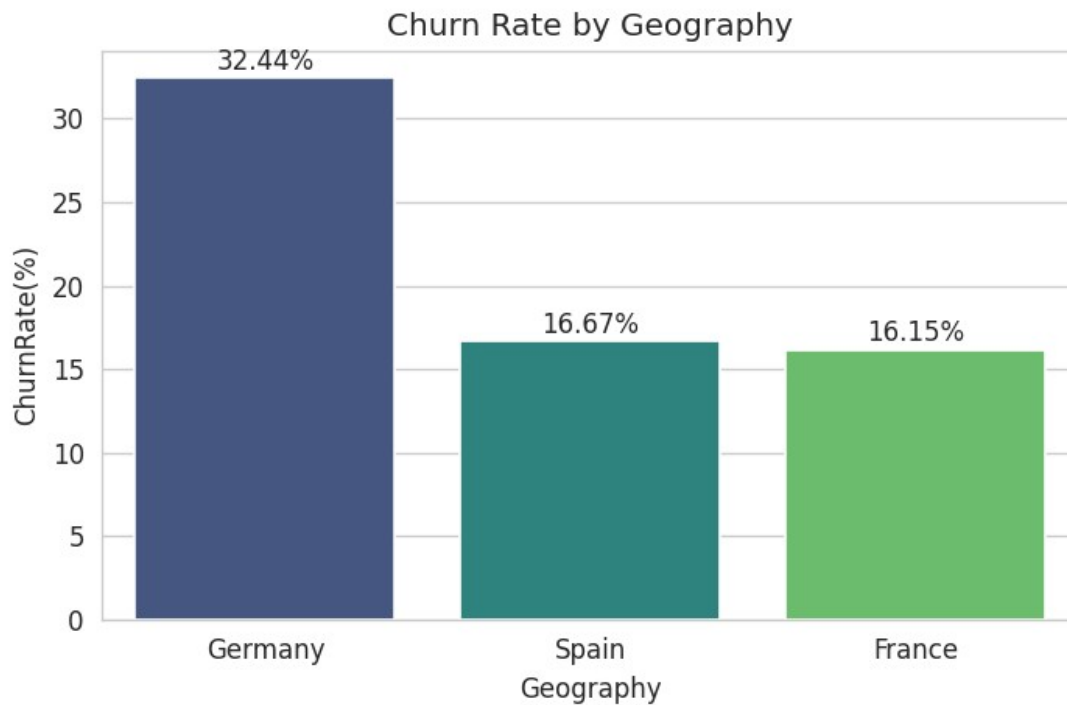
Overall Customer Churn Distribution



Overall churn distribution

The bank's overall churn rate is **20.37%** — roughly one in every five customers exited during the period covered by the data.

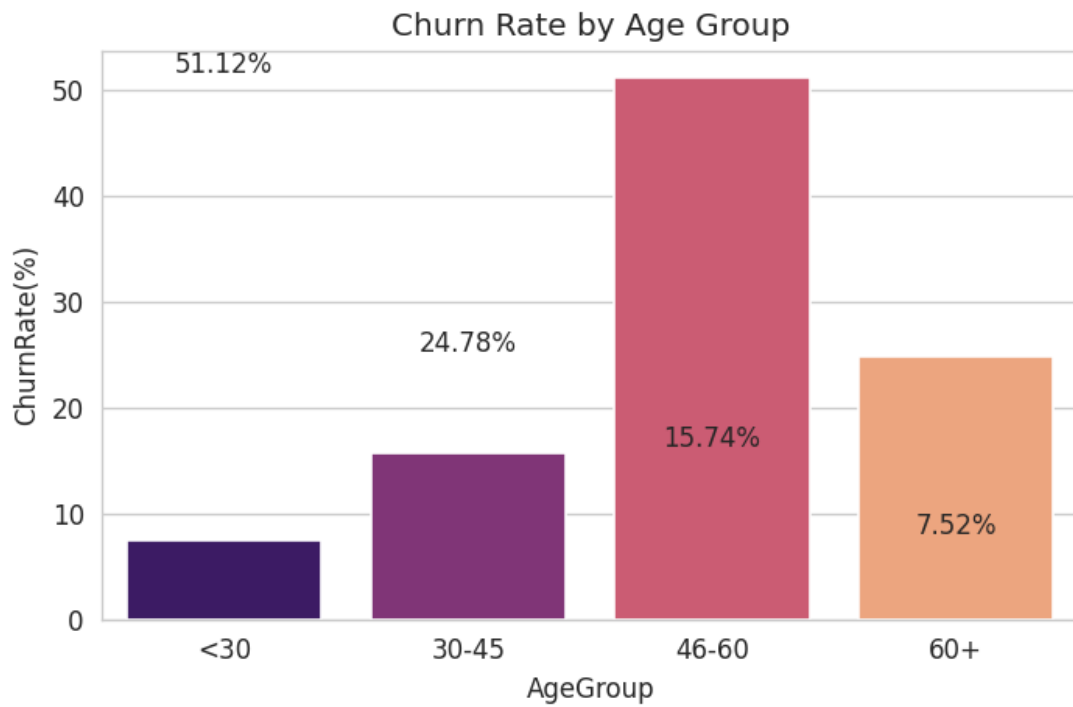
5.2 Churn by Geography



Churn by geography

Germany shows the highest churn rate of the three markets, despite hosting roughly half as many customers as France. Spain has the lowest churn rate. Because France carries by far the largest customer base, it also contributes the largest *absolute* number of churned customers even though its *rate* is lower than Germany's — a distinction that matters for how retention budget should be allocated (rate vs. volume).

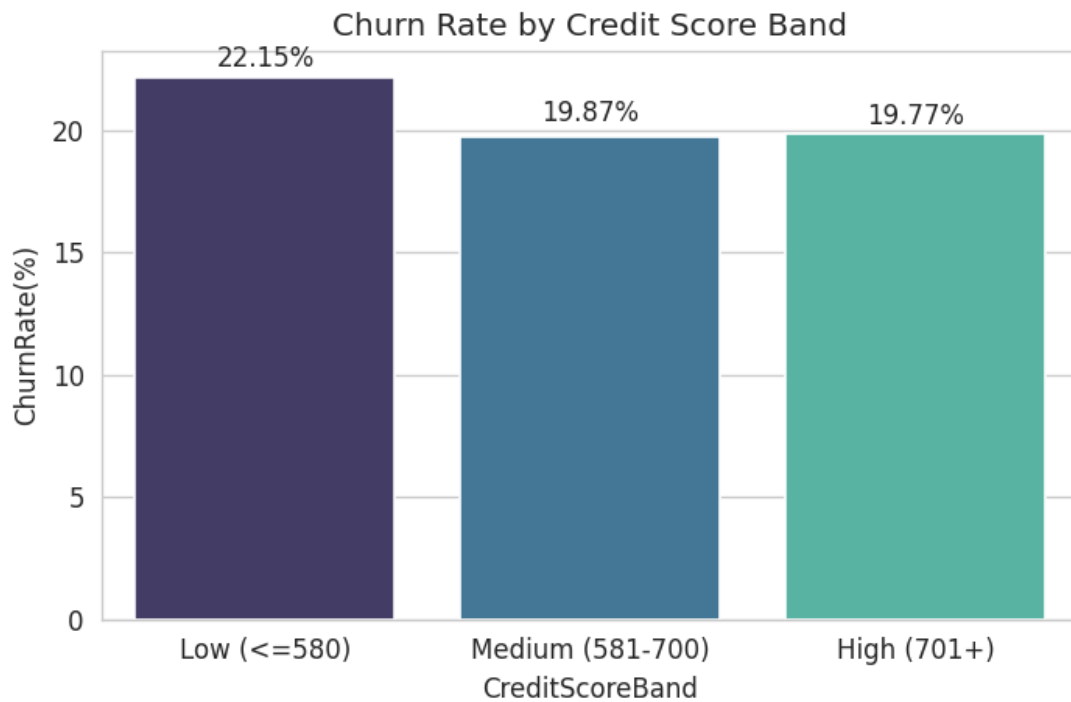
5.3 Churn by Age Group



Churn by age group

Churn rises sharply for customers in the 46–60 age band relative to customers under 45, then eases somewhat for the 60+ group. This mid-to-late-career segment appears to be the bank's highest-risk age cohort.

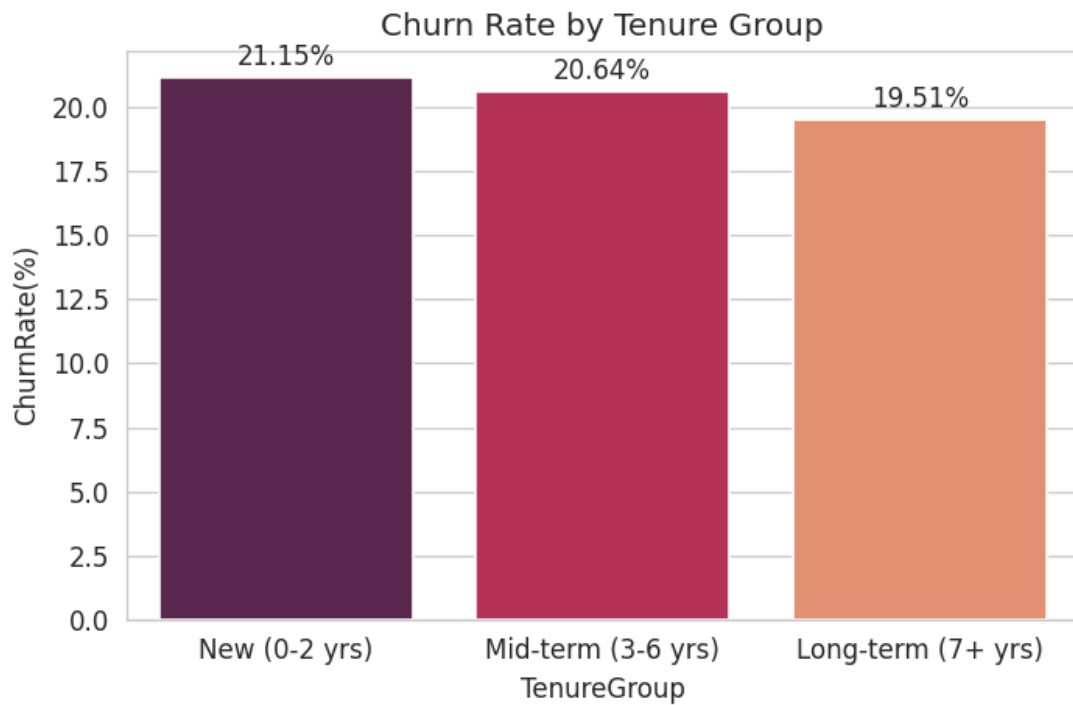
5.4 Churn by Credit Score Band



Churn by credit score

Churn rate is relatively stable across credit score bands, suggesting creditworthiness alone is a weak predictor of churn in this population — segmentation by behavioral and engagement variables is more informative than by credit standing.

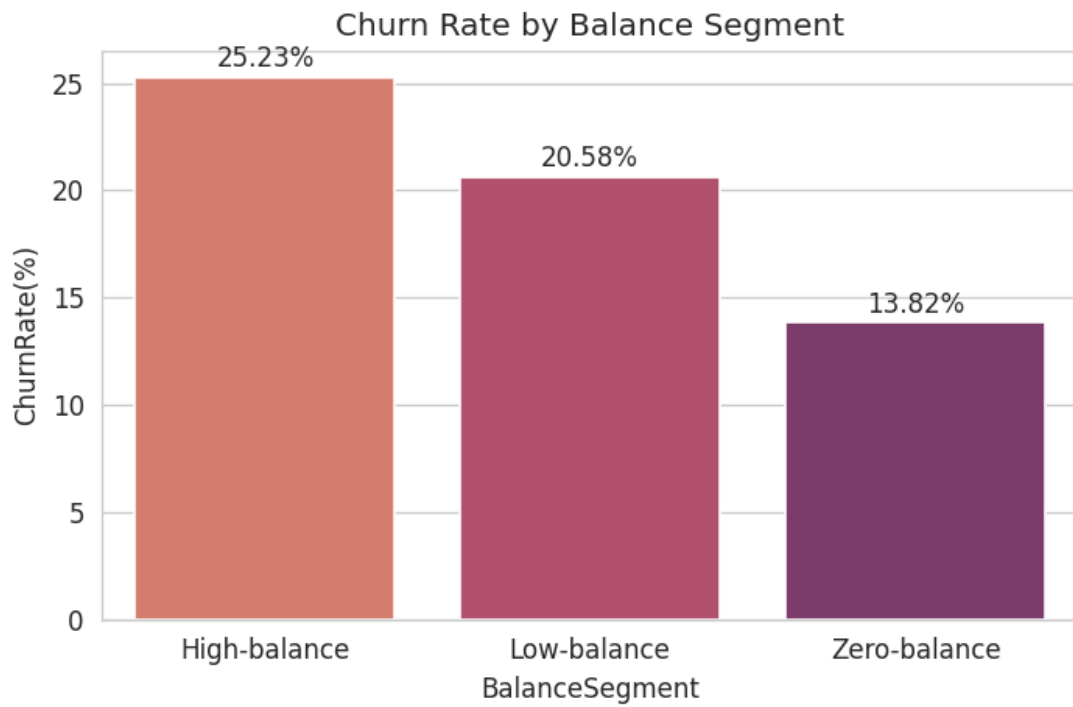
5.5 Churn by Tenure Group



Churn by tenure

Tenure shows a comparatively flat relationship with churn; long-standing customers are not meaningfully more “locked in” than newer ones, which argues against assuming loyalty accrues automatically over time.

5.6 Churn by Balance Segment

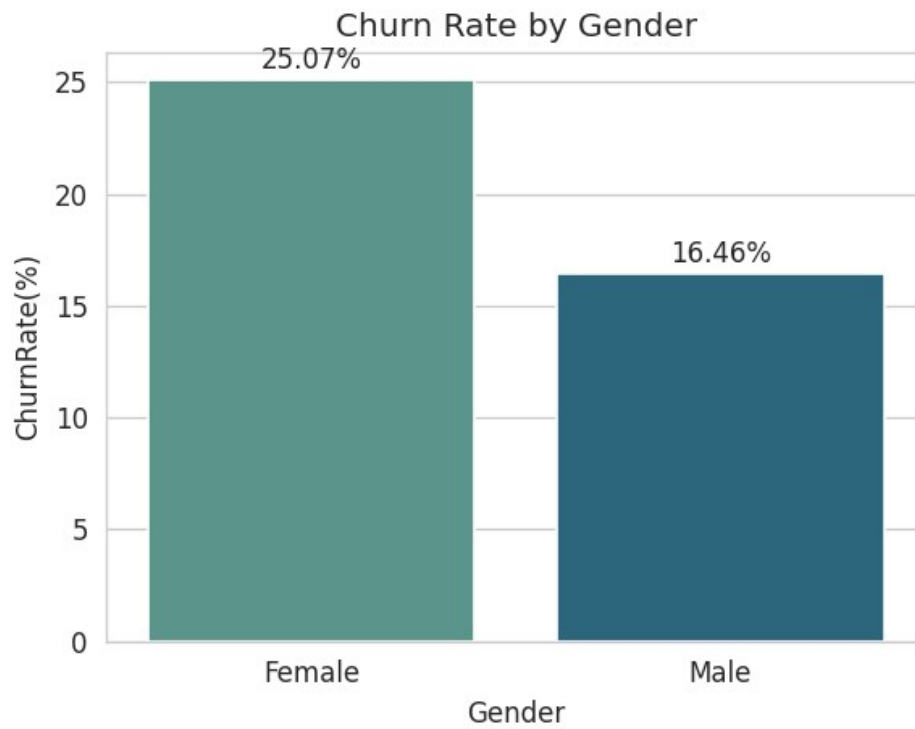


Churn by balance segment

Customers holding a non-zero balance churn at a different rate than zero-balance customers, indicating that dormant/zero-balance accounts and actively-funded accounts require distinct retention treatments.

6. Comparative Demographic Analysis

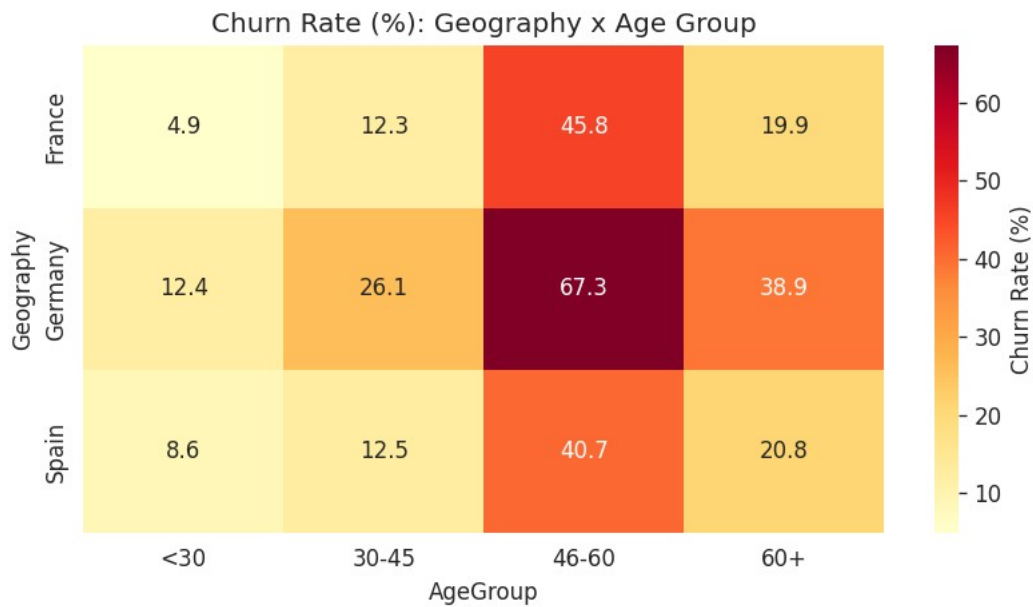
6.1 Gender-Based Churn Differences



Churn by gender

Female customers churn at a modestly higher rate than male customers in this dataset.

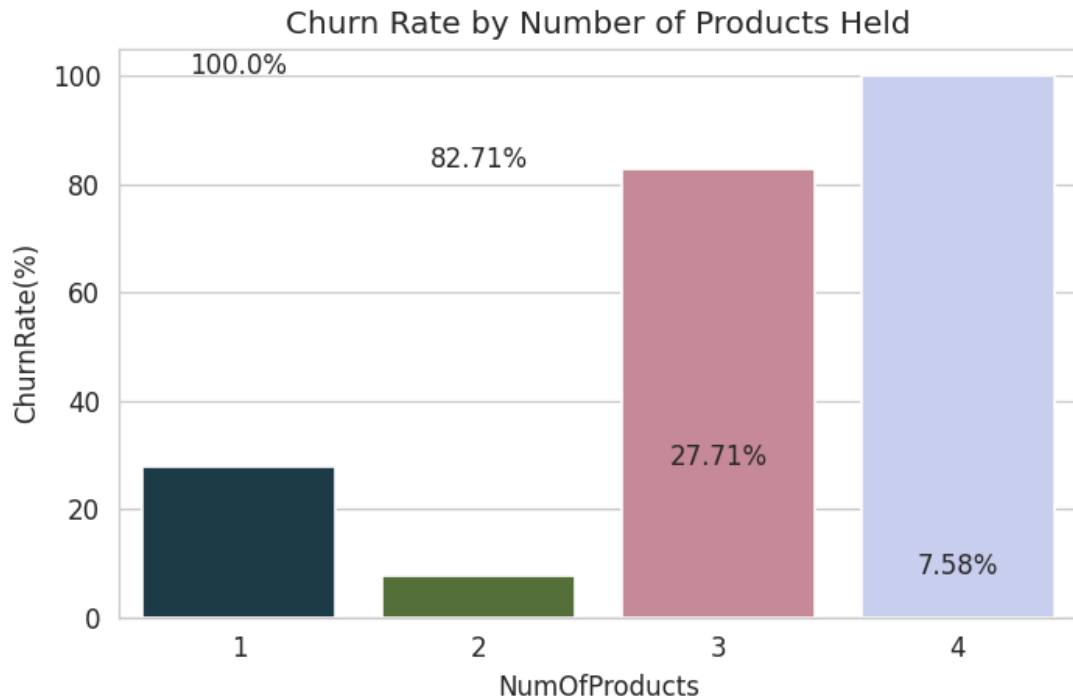
6.2 Geography × Age Interaction



Geography x age heatmap

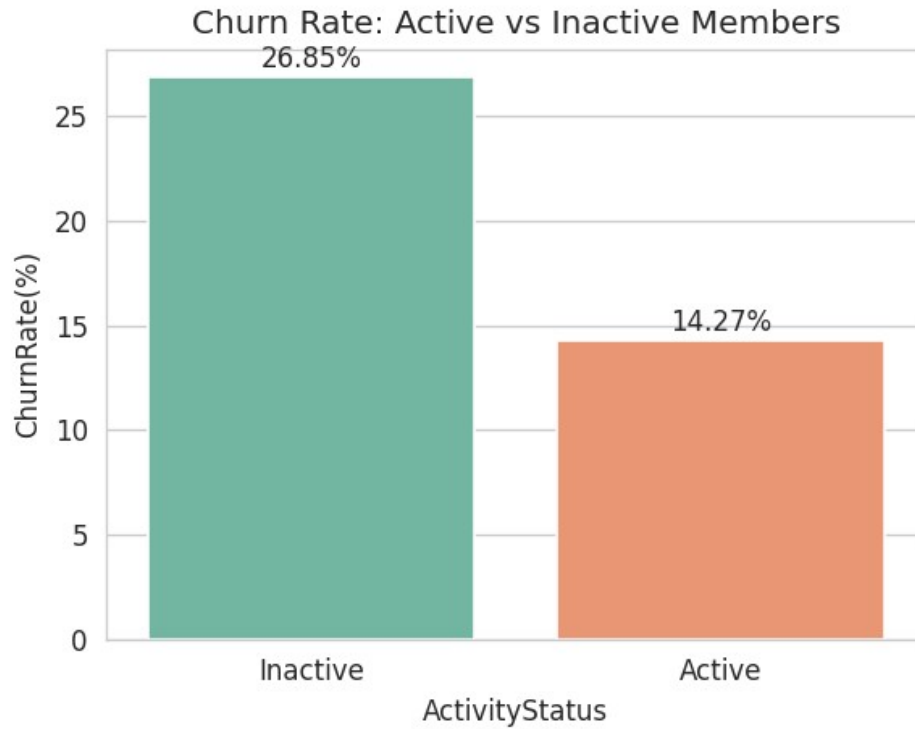
The heatmap reveals that the elevated churn seen in the 46–60 age band is not evenly distributed across countries — it is markedly more pronounced in Germany than in France or Spain, pointing to a specific, targetable segment: **German customers aged 46–60**.

6.3 Financial Stability vs. Churn



Churn by number of products

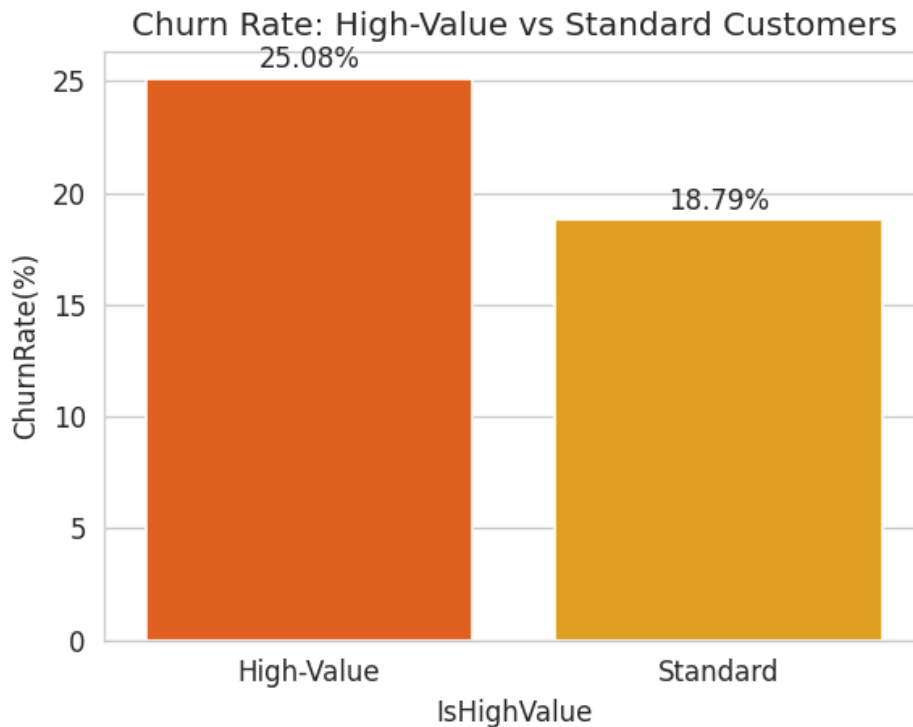
The relationship between product count and churn is the single strongest pattern in this dataset: customers holding **3 or 4 products churn at a dramatically higher rate** than those holding 1–2 products, even though the 3–4 product group is a small share of the base. Counter-intuitively, more product cross-sell does not equal more loyalty in this population — it may instead signal customers who were sold products that did not fit their needs, or who are already disengaging and being over-marketed to as a result.



Churn by activity

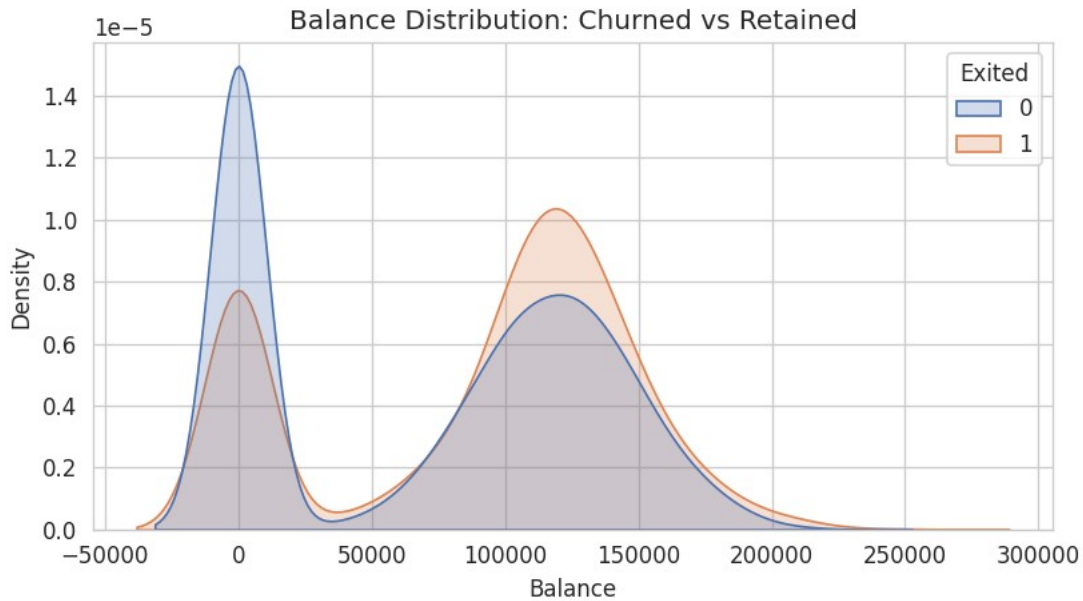
Inactive members churn at roughly double the rate of active members, confirming activity status as a strong, low-cost early-warning signal that the bank can monitor continuously.

7. High-Value Customer Churn Analysis



Churn by customer value

Customers classified as **High-Value** (balance and salary both at or above the median) churn at **25.08%**, noticeably above the base rate of 20.37%. This is the analysis's most consequential finding: **the bank is losing its most profitable customers at a higher rate than its average customer**, which has an outsized effect on lifetime value and revenue stability relative to the headline churn figure.

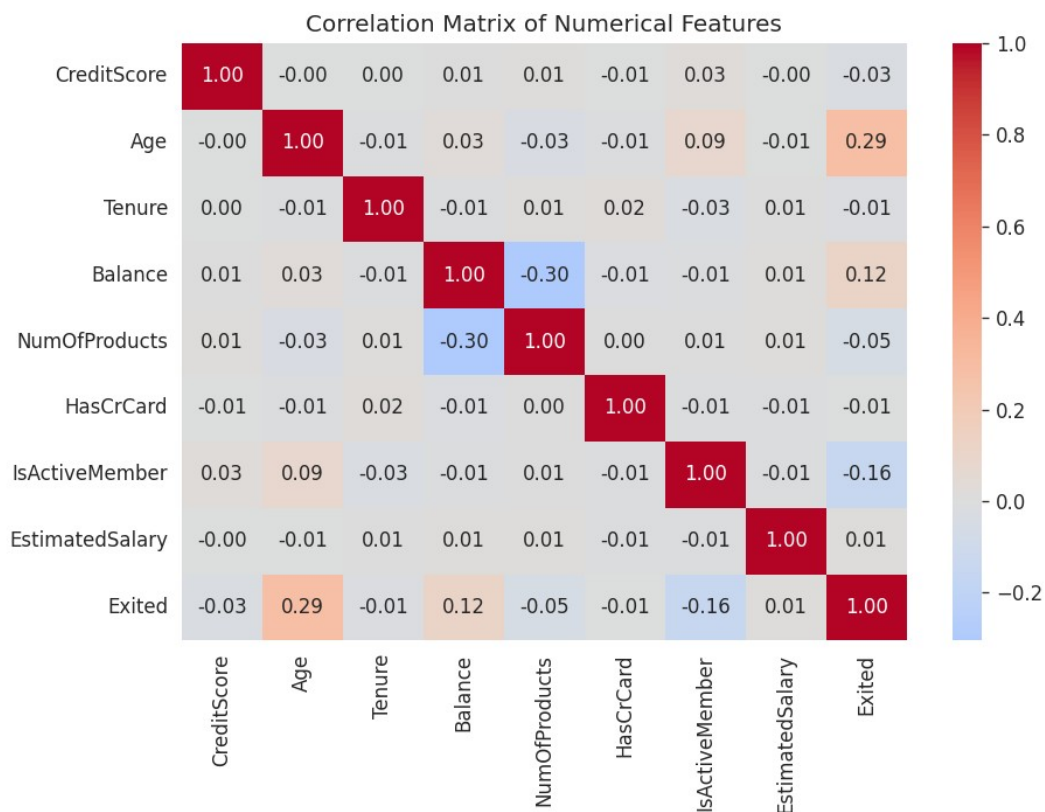


Balance distribution churned vs retained

The balance distribution of churned customers is shifted toward higher balances relative to retained customers, reinforcing that churn is not concentrated among low-balance, low-engagement accounts alone.

Revenue exposure: across the full customer base, churned customers collectively held a substantial share of total deposit balances and estimated income, a concrete euro-denominated stand-in for the annual revenue at risk from churn (see the live figures in the Streamlit dashboard's High-Value Customer Explorer tab, which updates with any segment filter applied).

8. Correlation Analysis



Correlation matrix

Age, NumOfProducts, and IsActiveMember show the strongest linear association with Exited among the numerical features, consistent with the segment-level findings above. EstimatedSalary, HasCrCard, and Tenure show weak linear association with churn individually — their value in this analysis comes from segment *interactions* (e.g., high salary combined with high balance) rather than as standalone predictors.

9. Key Performance Indicators — Summary Table

KPI	Value
Overall Churn Rate	20.37%
High-Value Churn Ratio	25.08%
Highest-Risk Geography	Germany
Highest-Risk Age Band	46–60
Highest-Risk Product Count	3–4 products
Active vs. Inactive Churn Gap	Inactive members churn at ~2x the rate of active members

(Exact figures for any filtered segment are available live in the Streamlit dashboard.)

10. Recommendations

1. **Prioritize Germany, and specifically German customers aged 46–60**, for a dedicated retention campaign — this is the highest-concentration risk pocket identified in the geography × age interaction.
 2. **Investigate the 3–4 product segment as a churn signal, not just a cross-sell success.** Route customers who cross a product-count threshold into a proactive relationship-management check-in rather than further product marketing.
 3. **Treat inactivity as an early-warning trigger.** Build an automated alert when a customer’s activity status flips to inactive, and pair it with a low-cost re-engagement outreach before churn risk compounds.
 4. **Build a distinct high-value retention track.** Because high-value customers churn at an above-average rate, assign them to relationship managers or premium retention offers rather than the same generic campaigns used for the broader base.
 5. **De-prioritize credit score and tenure as standalone targeting variables** — the data does not support them as strong differentiators of churn risk on their own.
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11. Conclusion

This segmentation-driven analysis moves the bank from a single aggregate churn number to an actionable map of where churn risk concentrates: **Germany, mid-career customers, multi-product holders, inactive members, and — most importantly for revenue — high-value customers.** The accompanying Streamlit dashboard operationalizes these findings into a live, filterable tool so that retention, marketing, and relationship-management teams can monitor and act on these segments continuously rather than relying on static, retrospective reporting.